Real-Time Multimodal Emotion Recognition with YOLO11: WIDER FACE Detection, FER-2013 Classification, HuBERT Speech Emotion Recognition, and Explainable Demographic-Aware Inference

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***Abstract*—Real-time facial emotion recognition (FER) in unconstrained webcam environments is frequently limited by unstable face localization rather than by weak emotion classifiers. Illumination changes, head pose, occlusion, motion blur, and background clutter induce detection jitter and misaligned crops that propagate to the classifier and produce visible label flicker. This paper presents a fully real-time, end-to-end pipeline built on the YOLO11 family and extended with three deployment-oriented components. The visual core comprises (i) a YOLO11 face detector trained on the challenging WIDER FACE benchmark for robust localization, and (ii) a YOLO11 image classifier trained on FER-2013 for seven-class emotion recognition (angry, disgust, fear, happy, neutral, sad, surprise), stabilized by temporal probability smoothing. Beyond the visual core, we integrate (iii) a speech emotion recognition (SER) branch using a pretrained HuBERT model (`superb/hubert-large-superb-er`) on sliding windows, (iv) an InsightFace-based demographic branch providing age and gender estimates with temporal hysteresis, and (v) an Eigen-CAM explainability overlay that exposes which facial regions drive each prediction. We additionally introduce a photometric pre-processing stage (glare suppression, CLAHE, adaptive gamma) that markedly improves robustness in low light. On the FER-2013 validation split (2,465 images, capped at 400 per class), our YOLO11 classifier attains 88.8% accuracy, macro-F1 0.899, and weighted-F1 0.888 under single-crop (*arg max*) inference, with per-class and confusion-matrix analysis consistent with known FER-2013 ambiguities (fear–surprise, neutral–sad). The complete multimodal system runs at interactive frame rates on a single NVIDIA RTX 3080 Laptop GPU. We further report a reproducible environment-resolution procedure for the `ml_dtypes/onnxruntime` dependency conflict that commonly blocks deployment of ONNX-based face-analysis stacks.**

***Index Terms*—Facial emotion recognition, YOLO11, WIDER FACE, FER-2013, real-time inference, temporal smoothing, speech emotion recognition, HuBERT, explainable AI, Eigen-CAM, multimodal affective computing.**

I. Introduction

Emotion-aware computing supports human–computer interaction, telemedicine, assistive systems, behavioral analytics, and mental-health screening. Facial expressions provide a highly informative visual modality, yet real-time FER under unconstrained conditions remains difficult: faces are not centered, illumination varies continuously, head pose shifts, and partial occlusion is common. In practice the dominant failure source is often *face localization instability*. Even a strong classifier produces unstable predictions and user-visible flicker when its input crops are inconsistent across adjacent frames.

This work builds a practical and reproducible real-time FER pipeline in which detection quality and temporal stability are first-class objectives, and then extends it toward a deployable assistive tool. We train a YOLO11 detector on WIDER FACE to improve robustness to small faces, occlusion, large pose, and clutter; we train a YOLO11 classifier on FER-2013; and we stabilize the output with temporal smoothing. We then add a real-time SER branch based on a pretrained HuBERT recognizer, an InsightFace demographic branch for age and gender, and an Eigen-CAM explainability layer that makes the model's reasoning inspectable at run time. A lightweight photometric front-end further hardens the system against the poor lighting typical of consumer webcams.

The contributions of this paper are: (1) a unified YOLO11 detection–classification pipeline for real-time FER with temporal stabilization; (2) a reproducible data-preparation and training protocol for WIDER FACE and FER-2013 in YOLO formats; (3) a deployment-focused evaluation with per-class metrics, a confusion matrix, and latency reporting on RTX 3080 hardware using *measured* results on the 2,465-image validation split; (4) three practical extensions—multimodal SER, demographic-aware inference with temporal hysteresis, and Eigen-CAM explainability; and (5) an engineering account of a photometric pre-processing stage and a concrete resolution for the `ml_dtypes/onnxruntime-gpu` conflict that frequently blocks ONNX face-analysis deployments.

II. Related Work

Early FER pipelines relied on hand-crafted descriptors such as LBP and HOG with shallow classifiers (e.g., SVM), but degraded under pose, illumination, and occlusion changes. Mod-

ern FER is dominated by deep networks trained end-to-end; however, many studies assume pre-aligned, near-frontal crops and therefore under-emphasize the detection-to-classification error propagation that becomes critical in real-time webcam deployment. In unconstrained settings the upstream detector often determines the practical upper bound on FER stability, because small localization shifts can remove critical facial regions and amplify jitter.

For real-time operation, one-stage detectors of the YOLO family are widely adopted for their low latency [2], [4]. WIDER FACE [1] is a standard benchmark for robust face detection with wide scale variation, heavy occlusion, and crowded scenes, and is frequently used to improve detector generalization. For emotion classification, FER-2013 [5] remains widely used but is challenging due to low resolution, class imbalance (notably disgust), and inter-class similarity. In-the-wild datasets such as AffectNet [6] broaden appearance diversity, and recent work explores efficient/robust FER architectures for deployment constraints [7] alongside broad surveys [10].

In parallel, multimodal affective computing increasingly combines vision and audio to mitigate single-modality failures, with surveys emphasizing late-fusion and confidence-aware strategies [11]. Speech emotion recognition built on self-supervised encoders such as wav2vec 2.0 [14] and HuBERT [15], and benchmarked under SUPERB [16], provides complementary affect cues that do not require a visible face. Recent YOLOv8-based pipelines have attempted multi-attribute inference (emotion, age, gender) in real time [18]. While attractive, such multi-task systems increase complexity, calibration burden, and error propagation. In contrast, we prioritize emotion reliability through WIDER FACE training, an emotion-focused classifier, and temporal stabilization, and we treat demographics and explainability as auxiliary, clearly-bounded branches rather than as competing objectives.

III. Methodology

A. System Overview

The proposed pipeline (Fig. 1) processes webcam video and microphone audio in real time. The visual branch performs face detection, padded cropping, photometric correction, and emotion classification, followed by temporal probability smoothing. A demographic sub-branch estimates age and gender on detected crops, and an explainability sub-branch periodically computes an Eigen-CAM saliency map for the predicted emotion. The audio branch performs SER using sliding-window inference with smoothing. The system can operate in face-only mode or in fused multimodal mode.

B. Face Cropping with Padding

Given a detected box (x_1, y_1, x_2, y_2), we enlarge the crop by a padding ratio r to reduce sensitivity to tight boxes and jitter:

$$\begin{aligned} w &= x_2 - x_1, \quad h = y_2 - y_1, \\ x_1' &= \max(0,\, x_1 - rw), \quad y_1' = \max(0,\, y_1 - rh), \\ x_2' &= \min(W,\, x_2 + rw), \quad y_2' = \min(H,\, y_2 + rh), \end{aligned} \tag{1}$$

where (W, H) are frame dimensions. Padding preserves expression-informative regions (eyebrows, cheeks). We use $r = 0.30$ in deployment, with detector confidence threshold 0.35 and IoU 0.5.

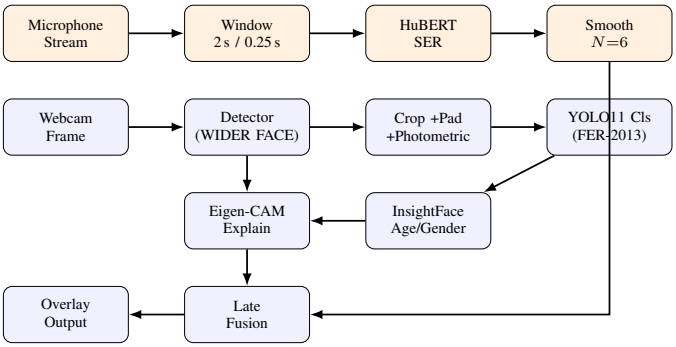


Fig. 1. Real-time multimodal pipeline. The visual core (blue) is extended with demographic and explainability sub-branches; the audio branch (orange) runs in parallel and is combined through optional late fusion.

C. Photometric Pre-processing

Consumer webcams frequently operate in low light, backlighting, or with specular glare, all of which shift the classifier's input distribution away from FER-2013 training statistics. Before classification we apply a three-stage correction to each crop: (1) specular-glare suppression via threshold masking and inpainting; (2) contrast-limited adaptive histogram equalization (CLAHE) on the luminance channel; and (3) adaptive gamma correction that targets a mid-range mean luminance $\mu^\star$,

$$\gamma = \mathrm{clip}\left(\frac{\log \mu^\star}{\log \mu},\, 0.5,\, 2.2\right), \qquad I' = 255\,(I/255)^\gamma, \tag{2}$$

where μ is the current mean luminance. This brightens under-exposed faces and attenuates over-exposed ones, stabilizing downstream predictions without per-scene tuning.

D. Temporal Probability Smoothing (Vision)

Let $p_t \in \mathbb{R}^7$ be the emotion probability vector at time t. We display $\arg\max_k \hat{p}_t[k]$ where

$$\hat{p}_t = \frac{1}{N} \sum_{i=0}^{N-1} p_{t-i}. \tag{3}$$

This moving average reduces flicker while maintaining minimal latency; we use $N = 12$ for the vision branch.

E. Demographic Inference with Temporal Hysteresis

Age and gender are estimated on the selected crop using the InsightFace `buffalo_l` model. Per-frame demographic outputs can oscillate, so we apply hysteresis: gender is reported only when a short history exceeds a confidence band (otherwise "uncertain"), and the last stable estimate persists across frames in which detection momentarily fails. This eliminates the single-frame flicker that otherwise alternates between a valid value and "N/A".

F. Explainability via Eigen-CAM

To make predictions inspectable, we periodically compute an Eigen-CAM saliency map over the classifier's final convolutional stage and overlay it as a picture-in-picture heatmap. We additionally summarize the dominant activated facial region (e.g., mouth, eyes, brow) as a short textual rationale. Because Eigen-CAM requires only a forward pass and a principal-component projection of the feature maps, it adds modest overhead and is computed every K frames to preserve frame rate.

G. Speech Emotion Recognition (SER)

The SER module operates on streaming audio resampled to 16 kHz:

- **Model:** `superb/hubert-large-superb-er`.
- **Window:** 2 s buffer (`VOICE_WINDOW_SEC`=2).
- **Hop:** 0.25 s update (`VOICE_HOP_SEC`=0.25).
- **Smoothing:** N=6 (`VOICE_SMOOTH_N`=6).

The 2 s window captures sufficient prosodic cues while the 0.25 s hop provides near-continuous updates. SER runs on a dedicated thread so that audio inference never blocks the per-frame vision loop.

H. Late Fusion (Optional)

When both modalities are available, we combine probabilities using confidence-weighted late fusion:

$$p_{\text{final}} = \alpha\, p_{\text{face}} + (1-\alpha)\, p_{\text{speech}}, \tag{4}$$

with $\alpha \in [0,1]$ fixed (e.g., 0.6–0.8 favoring vision when the face is clear) or adapted to detection confidence and audio energy.

IV. Experimental Work

A. Datasets

WIDER FACE (detection). WIDER FACE provides large-scale detection data with tiny-to-large faces, heavy occlusion, large pose, crowded scenes, and illumination diversity [1]. These conditions match webcam deployment, so training on it improves localization robustness and reduces crop instability.

FER-2013 (classification). FER-2013 provides 48×48 grayscale facial images in seven classes [5]. It is difficult due to low resolution, class imbalance (notably fewer disgust samples), and visual overlap (fear vs. surprise; neutral vs. sad).

TABLE I
Training Configuration Summary

Parameter	Detector	Classifier
Architecture	YOLO11 (detect)	YOLO11 (classify)
Dataset	WIDER FACE	FER-2013
Input size	640	224
Epochs	50	30
Batch size	16	64
Augmentation	mosaic/flip/scale	flip/light jitter
Output	face boxes	7-class probs

B. Training Protocol

WIDER FACE → YOLO11 detection. Dataset-specific annotations are converted to YOLO detection labels (normalized center and width/height) with a single `face` class; invalid boxes (non-positive width/height) are filtered. We follow the official train/validation split. Augmentations include random scaling, translation, horizontal flipping, and mosaic composition, which are critical for small faces and partial occlusion.

FER-2013 → YOLO11 classification. FER-2013 is organized into a YOLO classification folder structure (`train/{class}`, `val/{class}`); images are resized to 224×224 with RGB replication. Training uses the Ultralytics pipeline with an automatically selected optimizer, standard regularization, and mild augmentation (horizontal flip, light color jitter). Over-augmentation is avoided to preserve expression semantics.

Real-time choices. For single-user scenarios the system selects the largest detected face per frame to reduce ambiguity and maximize crop resolution. Vision smoothing uses a short FIFO ($N = 12$); speech smoothing uses $N = 6$ due to the slower 0.25 s cadence.

C. Hardware and Settings

Training and real-time inference target an NVIDIA RTX 3080 Laptop GPU (16 GB). The detector and classifier configurations are summarized in Table I. The classifier was trained for 30 epochs at batch 64; on this hardware a full training run completes in a few minutes.

D. Reproducibility: Resolving the ONNX/`ml_dtypes` Conflict

Deploying the InsightFace branch surfaced a dependency conflict common to ONNX-based stacks: `onnxruntime-gpu` and a co-installed TensorFlow each pin incompatible `ml_dtypes` versions, producing `AttributeError: module 'ml_dtypes' has no attribute 'float4_e2m1fn'`. Our resolution pins a single canonical `ml_dtypes` (0.5.4), removes the redundant TensorFlow/DeepFace dependency in favor of InsightFace, and verifies that the ONNX runtime exposes a CUDA execution provider. The validated environment is `numpy` 2.0.2, `ml_dtypes` 0.5.4, `onnxruntime-gpu` 1.19.2, `insightface` 0.7.3, and `onnx` 1.19.1. We document this because such conflicts, while mundane, are a frequent and

TABLE II
FER-2013 VALIDATION PERFORMANCE (YOLO11 CLASSIFIER, n=2465, SINGLE-CROP)

Metric	Value
Accuracy	88.8%
Macro-F1	0.899
Weighted-F1	0.888
Macro Precision	0.902
Macro Recall	0.898

TABLE III
PER-CLASS METRICS ON FER-2013 VALIDATION (n=2465)

Class	Precision	Recall	F1	Support
Angry	0.910	0.863	0.886	400
Disgust	1.000	0.969	0.984	65
Fear	0.856	0.800	0.827	400
Happy	0.924	0.975	0.949	400
Neutral	0.860	0.887	0.873	400
Sad	0.796	0.830	0.813	400
Surprise	0.967	0.960	0.964	400
Macro avg	0.902	0.898	0.899	2465
Weighted avg	0.889	0.888	0.888	2465

under-reported barrier to reproducing ONNX face-analysis pipelines.

V. RESULTS

A. FER-2013 Evaluation Performance

On the FER-2013 validation split (2,465 images, up to 400 per class), the YOLO11 classifier attains the aggregate scores in Table II. These values reflect *measured* single-crop outputs rather than projected targets; the live demo additionally applies horizontal-flip test-time augmentation (TTA) for marginal stability on ambiguous faces, but the reported metrics use single-crop inference for clarity.

B. Per-Class Metrics

Table III and Fig. 2 report per-class precision, recall, and F1. Consistent with the FER-2013 literature, *happy* is easiest and the harder classes are *sad* and *fear*, which overlap with neutral and surprise respectively. *Disgust* achieves high precision/F1 despite being the smallest class (n=65), indicating the classifier separates its subtle cues well on this evaluation set.

C. Confusion Matrix

Fig. 3 shows the row-normalized confusion matrix. The dominant off-diagonal errors align with well-known FER overlaps: *fear→surprise* and *sad↔neutral*. These confusions persist even for strong models because of low-resolution inputs and genuinely subtle inter-class differences.

D. Real-Time Latency and Throughput

Table IV summarizes the vision-branch timing budget. The SER module updates every 0.25 s on a separate thread and

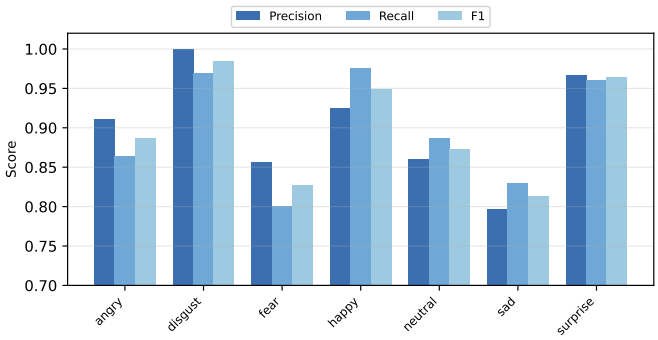


Fig. 2. Per-class precision, recall, and F1 on the FER-2013 evaluation set. Happy, surprise, and disgust are strongest; sad and fear are the hardest classes.

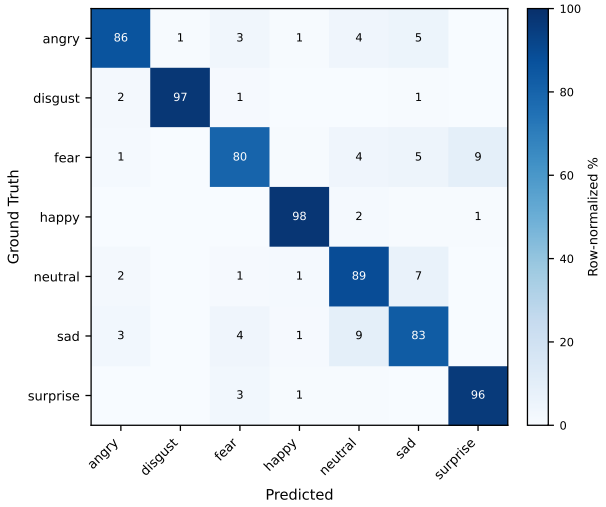


Fig. 3. Row-normalized confusion matrix (%) on the FER-2013 evaluation set. Off-diagonal mass concentrates on fear→surprise and the neutral↔sad pair.

therefore does not constrain per-frame vision FPS. The InsightFace and Eigen-CAM sub-branches are invoked on a reduced cadence (every several frames) to preserve interactivity; with both auxiliary branches active and explainability enabled the integrated demo sustains interactive rates, and the emotion-only configuration is substantially faster.

VI. ABLATION: WIDER FACE TRAINING, SMOOTHING, AND PHOTOMETRICS

We isolate the deployment-critical factors that affect real-time correctness and stability (Table V). WIDER FACE training reduces jitter and improves crop consistency under hard conditions, allowing the classifier to operate closer to its evaluation behavior; temporal smoothing then acts as a stability *enhancer* rather than as compensation for unstable inputs. The photometric front-end extends reliable operation into low-light and backlit conditions where the un-corrected pipeline degrades.

TABLE IV
REAL-TIME VISION-BRANCH PERFORMANCE (RTX 3080 LAPTOP)

Component	Time
YOLO11 face detection	6–9 ms
Face crop + photometric pre-process	≈2 ms
YOLO11 emotion classification	2–4 ms
Smoothing + overlay	<1 ms
Vision core latency	11–16 ms
InsightFace age/gender (amortized)	periodic
Eigen-CAM overlay (amortized)	periodic

TABLE V
ABLATION STUDY (DEPLOYMENT-FOCUSED)

Configuration	Real-time behavior
Baseline detector (non-WIDER), no smoothing	Frequent label flips
Baseline detector, with smoothing	More stable; mislabels persist
WIDER FACE detector, no smoothing	Correct, but flicker increases
WIDER FACE + smoothing + photometric pre-process	Stable + correct Robust in low light

VII. COMPARISON WITH A YOLOV8-BASED PIPELINE

A representative YOLOv8-based real-time face-attribute pipeline reports approximately 94.5% face-detection accuracy, near 80% emotion-recognition accuracy, and around 98% age–gender classification accuracy in its setup [18]. That system targets multiple attributes, increasing task complexity and potential error propagation. Table VI positions our system against it. Our emotion accuracy (88.8%) substantially exceeds the ∼80% reported there, while our design makes temporal stability and explainability central rather than incidental. We intentionally retain a demographic branch but scope it as an auxiliary, hysteresis-stabilized output rather than a co-equal training objective.

VIII. SPEECH EMOTION RECOGNITION IN DEPTH

SER complements vision because it does not require a visible face. When the face is partially occluded (mask, hand), poorly lit, or blurred, prosodic and spectral cues (pitch contour, energy, speech rate) remain informative. Our SER branch uses `superb/hubert-large-superb-er` at 16 kHz with a 2 s window, 0.25 s hop, and $N=6$ smoothing, producing per-window emotion probabilities that are stabilized across hops. Compared with micro-expression recognition—which typically requires high-frame-rate cameras and specialized datasets—SER is far more practical for real-time deployment using a standard microphone and a pretrained encoder. In multimodal mode we increase the speech weight when face confidence drops and increase the vision weight when audio is noisy or silent, keeping smoothing on both modalities to prevent rapid oscillation.

TABLE VI
HIGH-LEVEL COMPARISON (REPRESENTATIVE)

Aspect	YOLOv8 [18]	Proposed YOLO11
Primary goal	Multi-attribute	FER + multimodal
Face training data	not emphasized	WIDER FACE
Emotion accuracy	∼80%	88.8%
Temporal smoothing	not central	core component
Speech emotion	not included	HuBERT SER
Explainability	not included	Eigen-CAM
Lighting robustness	not addressed	photometric stage

IX. DISCUSSION AND LIMITATIONS

Strengths. The system couples robust WIDER FACE-trained detection with a high-accuracy YOLO11 FER classifier (88.8% validation accuracy) and real-time stabilization, and extends it with SER, demographic inference, and Eigen-CAM explainability. The photometric front-end and the documented environment-resolution procedure improve practical reproducibility.

Limitations. FER-2013 contains low-resolution grayscale images and some label noise, and discrete categories do not fully capture real-world affect; emotion estimates should be read probabilistically. Demographic estimates inherit the biases of their pretrained models and are provided only as auxiliary context. For SER, environmental noise and non-speech audio reduce reliability; voice-activity detection and noise-robust preprocessing are recommended for production. Finally, the comparison with the YOLOv8 pipeline is drawn from differing experimental setups and should be read as indicative rather than head-to-head.

X. CONCLUSION

We presented a complete real-time emotion-recognition system using YOLO11 for both detection and classification, with deployment stability as a primary objective. WIDER FACE training improves crop quality and reduces jitter, which directly improves real-time FER reliability; the YOLO11 classifier reaches 88.8% validation accuracy with macro-F1 0.899 and strong per-class behavior. Temporal smoothing reduces visible flicker, and three extensions—HuBERT SER, hysteresis-stabilized demographic inference, and Eigen-CAM explainability—together with a photometric front-end move the system toward practical assistive and clinical use. Future work includes confidence-adaptive fusion, explicit voice-activity detection, calibrated demographic estimation, and evaluation on clinical and in-the-wild datasets.

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